Also, there is a simple geometrical way to calculate $\mathbf{a} \cdot \mathbf{b}$, without having to calculate the components of **a** and **b**: $\mathbf{a} \cdot \mathbf{b}$ is the product of the length of **a** and the length of **b** times the cosine of the angle between them. Why? Suppose that we choose a special coordinate system in which the $x$-axis lies along **a**; in those circumstances, the only component of **a** that will be there is $a_x$, which is of course the whole length of **a**. Thus Eq. (1.19) reduces to $\mathbf{a} \cdot \mathbf{b} = a_x b_x$ for this case, and this is the length of **a** times the component of **b** in the direction of **a**, that is, $b \cos \theta$:

$$\mathbf{a} \cdot \mathbf{b} = ab \cos \theta.$$

Therefore, in that special coordinate system, we have proved that $\mathbf{a} \cdot \mathbf{b}$ is the length of **a** times the length of **b** times $\cos \theta$. But *if it is true in one coordinate system, it is true in all*, because $\mathbf{a} \cdot \mathbf{b}$ is independent of the coordinate system; that is our argument.

What good is the dot product? Are there any cases in physics where we need it? Yes, we need it all the time. For instance, in Chapter 4* the kinetic energy was called $\frac{1}{2}mv^2$, but if the object is moving in space it should be the velocity squared in the $x$-direction, the $y$-direction, and the $z$-direction, and so the formula for kinetic energy according to vector analysis is

$$\text{K.E.} = \frac{1}{2}m(\mathbf{v} \cdot \mathbf{v}) = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2). \quad (1.22)$$

Energy does not have direction. Momentum has direction; it is a vector, and it is the mass times the velocity vector.

Another example of a dot product is the work done by a force when something is pushed from one place to the other. We have not yet defined work, but it is equivalent to the energy change, the weights lifted, when a force **F** acts through a distance **s**:

$$\text{Work} = \mathbf{F} \cdot \mathbf{s}. \quad (1.23)$$

It is sometimes very convenient to talk about the component of a vector in a certain direction (say the vertical direction because that is the direction of gravity). For such purposes, it is useful to

* of *The Feynman Lectures on Physics*, vol. I.